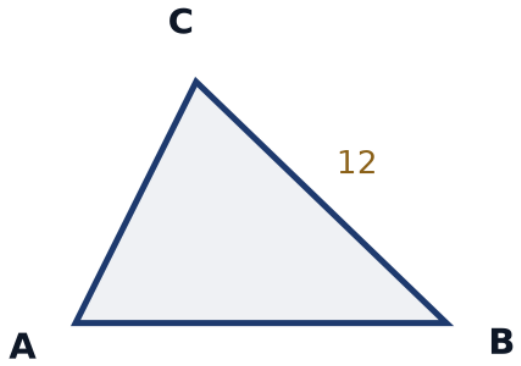
Law of Sines and Law of Cosines

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.04

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Law of Sines

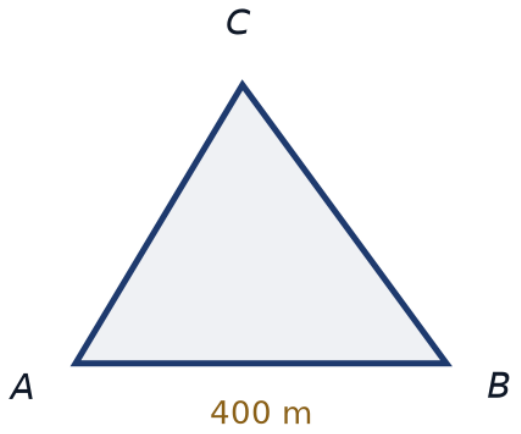
In triangle ABC, $A = 40°$, $B = 65°$, and $a = 12$. Find:



a angle C **b** side b (2 d.p.) **c** side c (2 d.p.)

2 [Foundational] • Target: Law of Sines

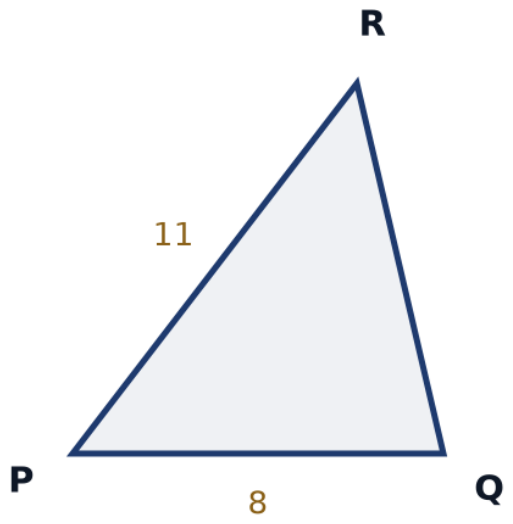
Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are $58°$ and $47°$. Find the distance from the balloon to the first observer (the one with the $58°$ sighting), to the nearest metre.



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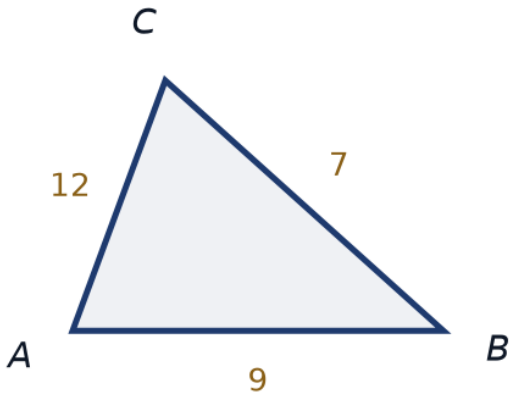
3 [Foundational] • Target: Law of Cosines

In triangle PQR, two sides of length 8 and 11 enclose an angle of $52°$. Find the length of the third side, to two decimal places.



4 [Foundational] • Target: Law of Cosines

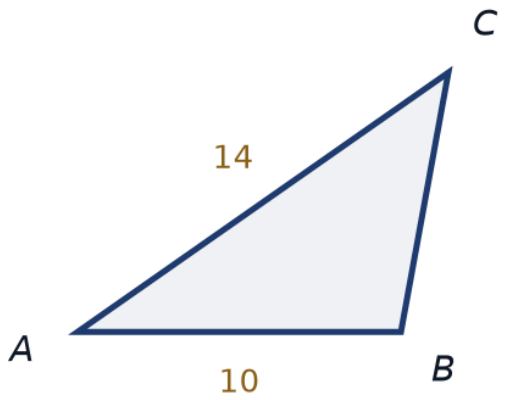
A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.



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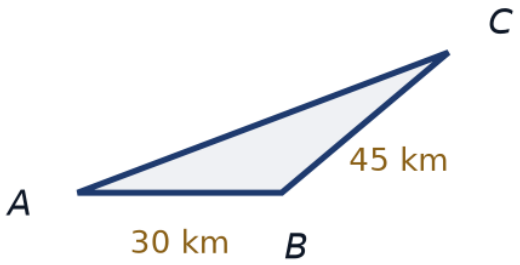
5 [Foundational] • Target: Law of Cosines

Find the area of a triangle with sides 10 and 14 enclosing an angle of $35°$, to two decimal places.



6 [Foundational] • Target: Law of Cosines

A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N\,50°\,E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.



7 [Foundational] • Target: The ambiguous case (SSA)

In triangle ABC, $A = 30°$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The ambiguous case (SSA)

In triangle DEF, $D = 40°$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

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9 [Intermediate] • Target: Choosing the right law

For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:

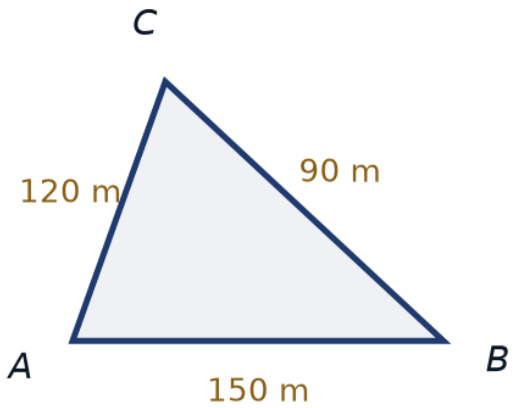
a Two angles and one side are known (AAS).

b Three sides are known (SSS).

c Two sides and the included angle are known (SAS).

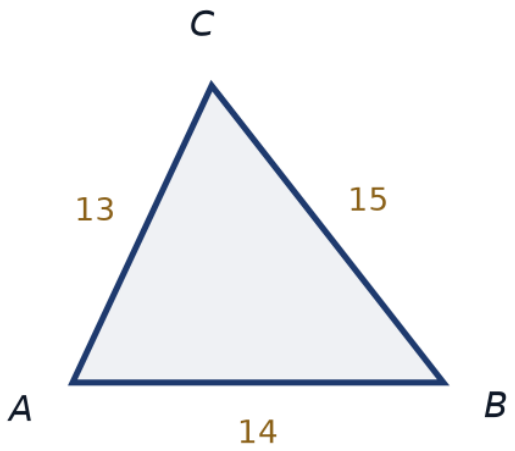
10 [Intermediate] • Target: Choosing the right law

A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.



11 [Intermediate] • Target: Heron's formula

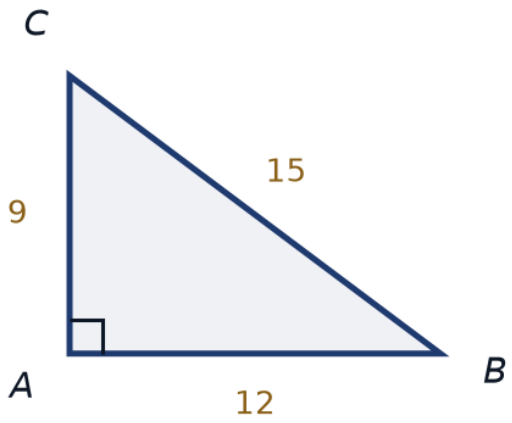
State Heron's formula for the area of a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$, then use it to find the area of a triangle with sides 13, 14, 15.



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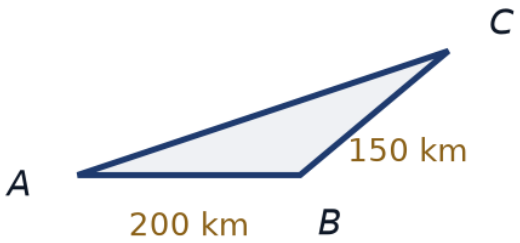
12 [Intermediate] • Target: Heron's formula

A triangular garden has sides 9 m, 12 m, and 15 m. Use Heron's formula to find its area, and verify it is a right triangle by checking the Pythagorean relationship.



13 [Intermediate] • Target: Navigation and bearing problems

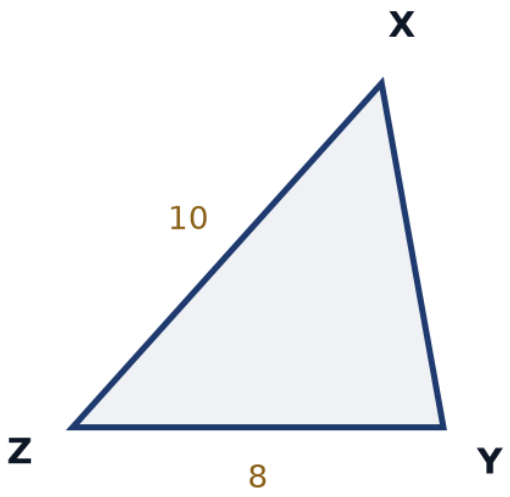
A plane flies on a bearing of $N\,60^\circ\,E$ for 200 km, then changes course to a bearing of $S\,80^\circ\,E$ for 150 km. Find the angle between the two legs of the flight, and set up (but do not evaluate) the Law of Cosines expression for the total displacement.



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14 **[Intermediate]** • **Target: Solving oblique triangles from a diagram**

In triangle XYZ, $x = 8$, $y = 10$, and angle $Z = 48°$ (between the given sides). Find side z, then find angle X using the Law of Sines, to the nearest tenth of a degree.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

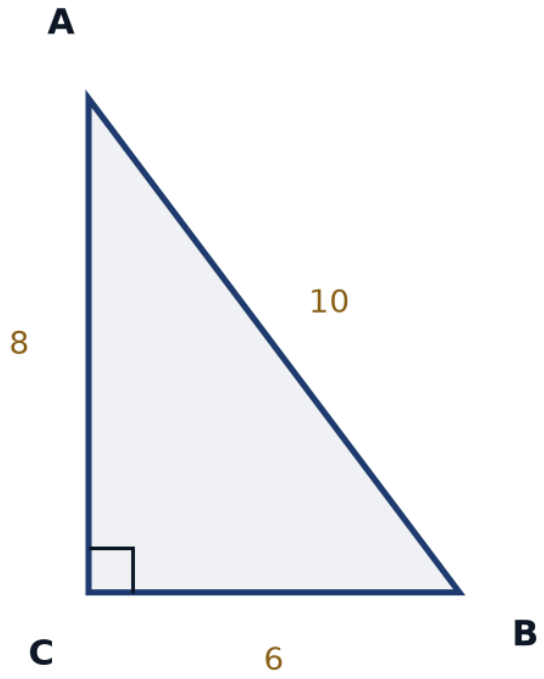
15 **[Challenging]** • **Target: Verifying triangle solutions with the angle sum** • **Cross-Lesson: 7.03 Exponential and Logarithmic Equations**

For the triangle solved above ($z \approx 7.55$, $X \approx 51.9°$, $Z = 48°$), find angle Y and verify all three angles sum to $180°$.

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16 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 7.04 Applications: Growth and Decay

A triangle has $a = 6$, $b = 8$, $c = 10$. Find angle C (opposite the longest side) two different ways: (1) using the Law of Cosines, and (2) recognizing this as a Pythagorean triple.



17 [Challenging] • Target: Choosing the right law • Cross-Lesson: 8.01 Rational Functions and Asymptotes

In ΔABC, $\angle A = 45^\circ$, $\angle B = 120^\circ$, and side $a = 11$. Find side b (Law of Sines).

18 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.02 Graphing Rational Functions

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side c. Round your answer to two decimal places.

19 [Challenging] • Target: Navigation and bearing problems • Cross-Lesson: 8.03 Introduction to Limits

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

20 [Challenging] • Target: Solving oblique triangles from a diagram • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find the angle between $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle -1, 1 \rangle$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Law of Sines

State the Law of Sines for any triangle ABC.

a. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

b. $a^2 = b^2 + c^2 - 2bc\cos A$

c. $a\sin A = b\sin B = c\sin C$

d. $\frac{a}{\cos A} = \frac{b}{\cos B}$

22 [Foundational] • Target: The Law of Sines

In ΔABC, $A = 30^\circ$, $B = 45^\circ$, and $a = 8$. Find the exact length of side b.

a. $8\sqrt{2}$

b. $4\sqrt{2}$

c. $8\sqrt{3}$

d. 16

23 [Foundational] • Target: The Law of Sines

Under which given information cases can the Law of Sines be directly applied to begin solving a triangle?

a. AAS or ASA

b. SAS or SSS

c. SSS only

d. Right triangles only

24 [Foundational] • Target: The Ambiguous Case (SSA)

In the ambiguous SSA case with acute angle A, what does $h = b\sin A$ represent?

a. The altitude from vertex C to side c

b. The perimeter of the triangle

c. The hypotenuse

d. The length of side a

25 [Foundational] • Target: The Ambiguous Case (SSA)

If $A = 30^\circ$, $b = 10$, and $a = 4$, how many triangles can be formed?

a. 0 (no triangle exists)

b. 1 triangle

c. 2 triangles

d. Infinitely many

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26 [Foundational] • Target: The Ambiguous Case (SSA)

If $A = 30^\circ$, $b = 10$, and $a = 7$, how many triangles can be formed?

a. 2 triangles

b. 1 triangle

c. 0 triangles

d. Cannot be determined

27 [Foundational] • Target: The Law of Cosines

State the Law of Cosines to solve for side c given sides a, b and included angle C.

a. $c^2 = a^2 + b^2 - 2ab\cos C$

b. $c^2 = a^2 + b^2 + 2ab\cos C$

c. $c = a\cos B + b\cos A$

d. $c^2 = a^2 + b^2 - ab\sin C$

28 [Foundational] • Target: The Law of Cosines

In ΔABC, $a = 5$, $b = 7$, and $C = 60^\circ$. Find the exact length of side c.

a. $\sqrt{39}$

b. $\sqrt{74}$

c. $\sqrt{49} = 7$

d. 6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Law of Cosines

In ΔABC, $a = 3$, $b = 5$, $c = 7$. Find the exact value of $\cos C$.

a. $-1/2$

b. $1/2$

c. $-1/5$

d. $-3/7$

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30 [Intermediate] • Target: Area of an Oblique Triangle

State the sine formula for the area of ΔABC given sides a and b and angle C.

a. $\text{Area} = \frac{1}{2}ab\sin C$

b. $\text{Area} = ab\sin C$

c. $\text{Area} = \frac{1}{2}ab\cos C$

d. $\text{Area} = \frac{a+b}{2}\sin C$

31 [Intermediate] • Target: Area of an Oblique Triangle

Find the exact area of ΔABC if $a = 6$, $b = 10$, and $C = 30^\circ$.

a. 15

b. 30

c. $15\sqrt{3}$

d. 7.5

32 [Intermediate] • Target: Area of an Oblique Triangle

Heron's formula states $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$. Find the area of a triangle with sides 5, 6, and 7.

a. $6\sqrt{6}$

b. $12\sqrt{3}$

c. 18

d. 14

33 [Intermediate] • Target: Area of an Oblique Triangle

In ΔABC, $\angle A = 120^\circ$, $\angle B = 50^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.



a. 9.6

b. 11.6

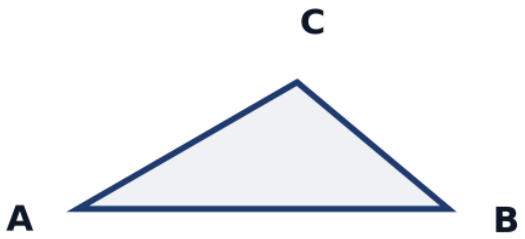
c. 5.0

d. 10.6

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34 [Intermediate] • Target: Law of Sines

In ΔABC, $\angle A = 30^\circ$, $\angle B = 40^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.



a. 12.9

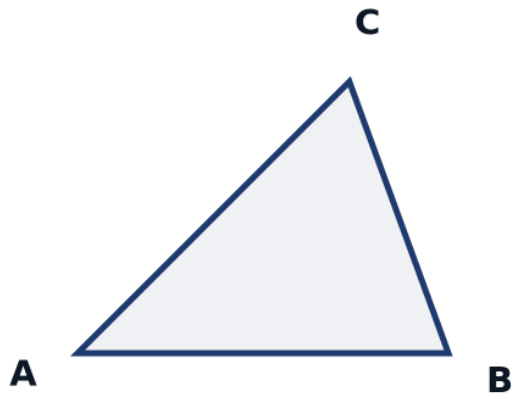
b. 11.9

c. 13.3

d. 13.9

35 [Intermediate] • Target: Law of Cosines

In ΔABC, $\angle A = 45^\circ$, $\angle B = 70^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.



a. 14.3

b. 13.3

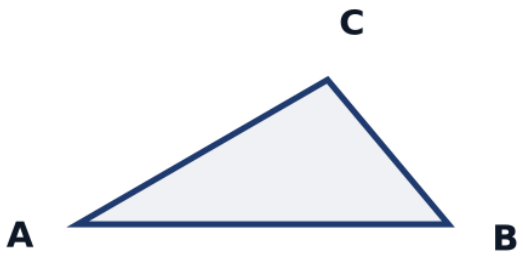
c. 12.3

d. 15.6

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36 [Intermediate] • Target: The ambiguous case (SSA)

In ΔABC, $\angle A = 30^\circ$, $\angle B = 50^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.



a. 19.4

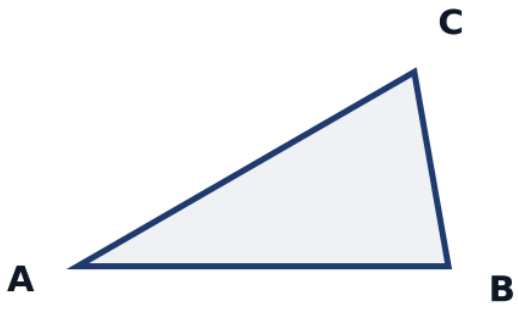
b. 20.0

c. 18.4

d. 17.4

37 [Intermediate] • Target: Choosing the right law

In ΔABC, $\angle A = 30^\circ$, $\angle B = 80^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.



a. 22.6

b. 32.0

c. 23.6

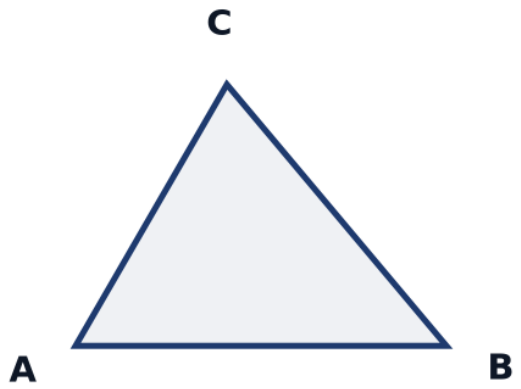
d. 24.6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.02 Graphing Rational Functions

In ΔABC, $\angle A = 60^\circ$, $\angle B = 50^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.



a. 7.8

b. 9.8

c. 8.3

d. 8.8

39 [Challenging] • Target: Navigation and bearing problems • Cross-Lesson: 8.03 Introduction to Limits

In ΔABC, $\angle A = 120^\circ$, $\angle B = 40^\circ$, and side $a = 8$. Find side b (to one decimal place) using the Law of Sines.



a. 6.9

b. 2.7

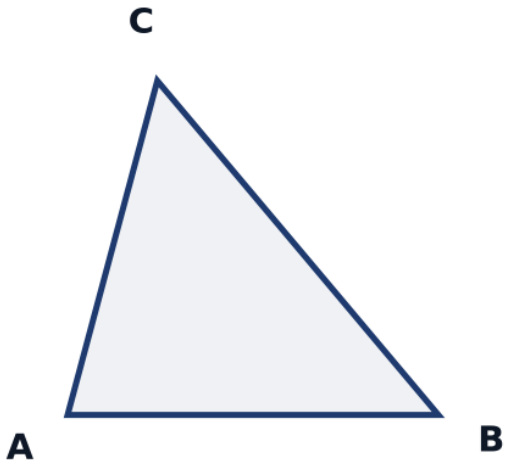
c. 5.9

d. 4.9

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40 [Challenging] • Target: Solving oblique triangles from a diagram • Cross-Lesson: 9.01 Arithmetic Sequences and Series

In ΔABC, $\angle A = 75^\circ$, $\angle B = 50^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.



a. 8.0

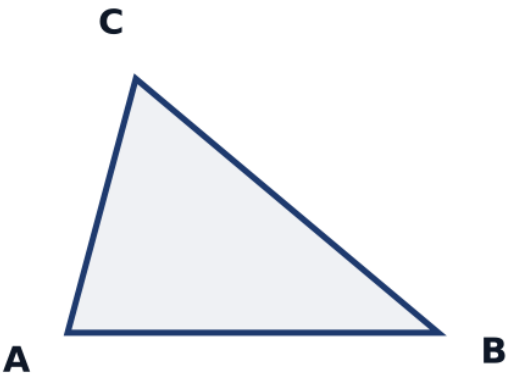
b. 9.5

c. 8.5

d. 10.5

41 [Challenging] • Target: Verifying triangle solutions with the angle sum • Cross-Lesson: 9.02 Geometric Sequences and Series

In ΔABC, $\angle A = 75^\circ$, $\angle B = 40^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.



a. 9.0

b. 8.0

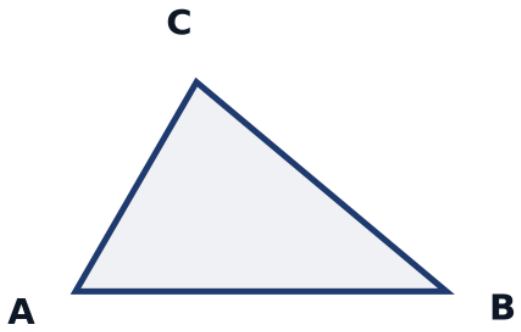
c. 7.0

d. 6.4

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42 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 9.03 The Binomial Theorem

In ΔABC, $\angle A = 60^\circ$, $\angle B = 40^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.



a. 9.9

b. 8.9

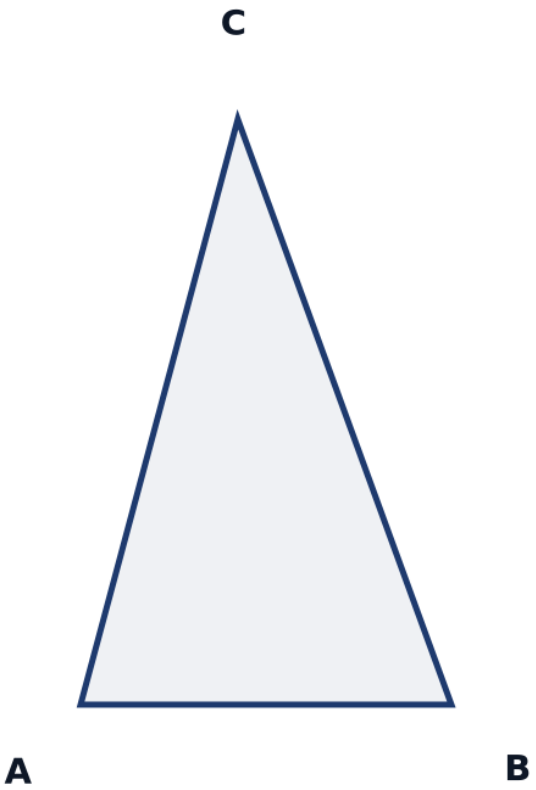
c. 7.9

d. 8.0

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43 [Challenging] • Target: The Law of Sines • Cross-Lesson: 1.01 Functions and Function Notation

In ΔABC, $\angle A = 75^\circ$, $\angle B = 70^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.



a. 10.7

b. 8.7

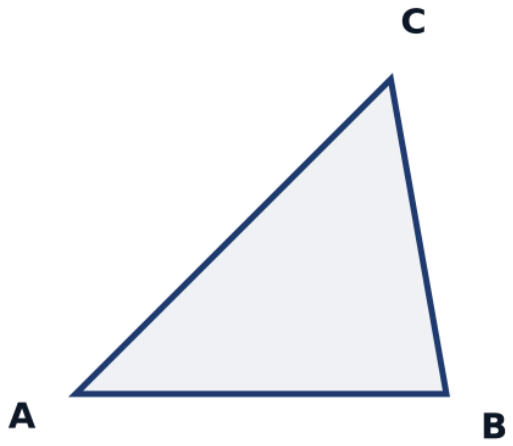
c. 9.7

d. 9.3

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44 [Challenging] • Target: The Ambiguous Case (SSA) • Cross-Lesson: 1.02 Domain and Range

In ΔABC, $\angle A = 45^\circ$, $\angle B = 80^\circ$, and side $a = 8$. Find side b (to one decimal place) using the Law of Sines.



a. 10.1

b. 12.1

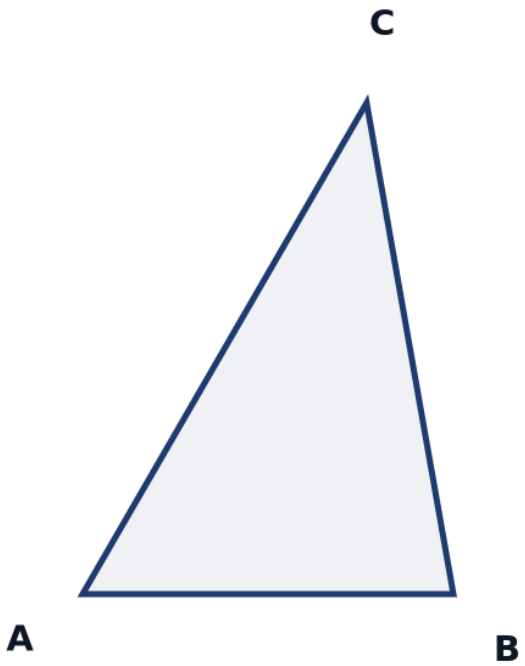
c. 14.2

d. 11.1

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45 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 1.03 Average Rate of Change

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.



a. 5.7

b. 6.7

c. 7.7

d. 4.7

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